MASS ANALYZERS

- Its function is to differentiate among ions according to their mass-to-charge ratio.
- There are a variety of mass analyzer designs.

1. Magnetic sector mass analyzers
2. quadrupole mass analyzers
3. time-of-flight (TOF),
4. ion trap, and
5. ion cyclotron resonance mass analyzers

scanning instruments

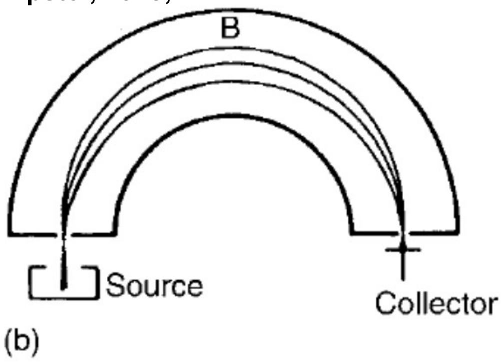
simultaneous transmission of all ions

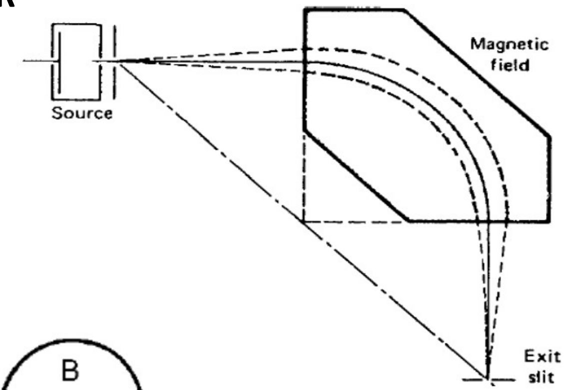
MAGNETIC AND ELECTRIC SECTOR INSTRUMENTS

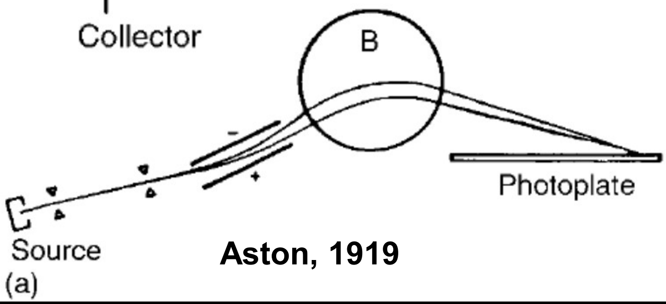
- An ion moving through a magnetic field B will follow a circular path with radius r.
- Changing B as a function of time allows ions of different m/z values to pass through the fixed radius flight tube sequentially.
- This scanning magnetic sector sorts ions according to their masses, assuming that all ions have a+1 charge and the same kinetic energy.
- The sector can have any apex angle, but 60^o and 90^o are common.
- A divergent beam of ions of a given m/z will be brought to a focus by passing through a sector shaped magnetic field.

NON-DISPERSIVE SECTOR ANALYZER

Dempster, 1918;



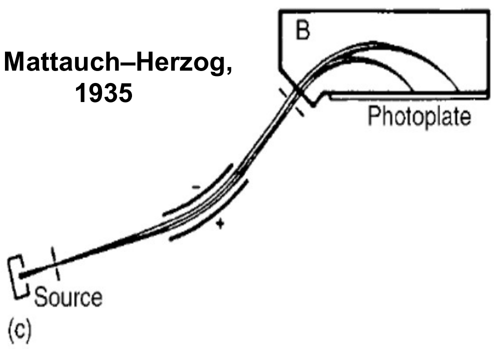


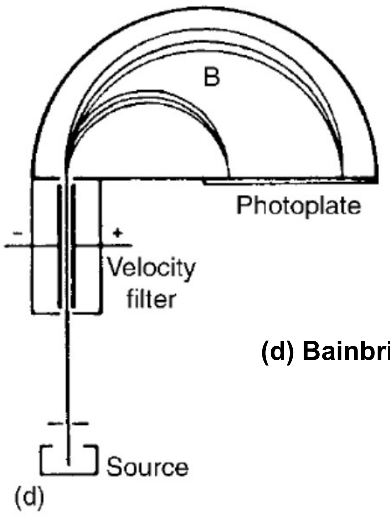


Aston, 1919

A dispersive magnetic sector mass analyzer does not use a flight tube with a fixed radius.

(c) Mattauch–Herzog, 1935





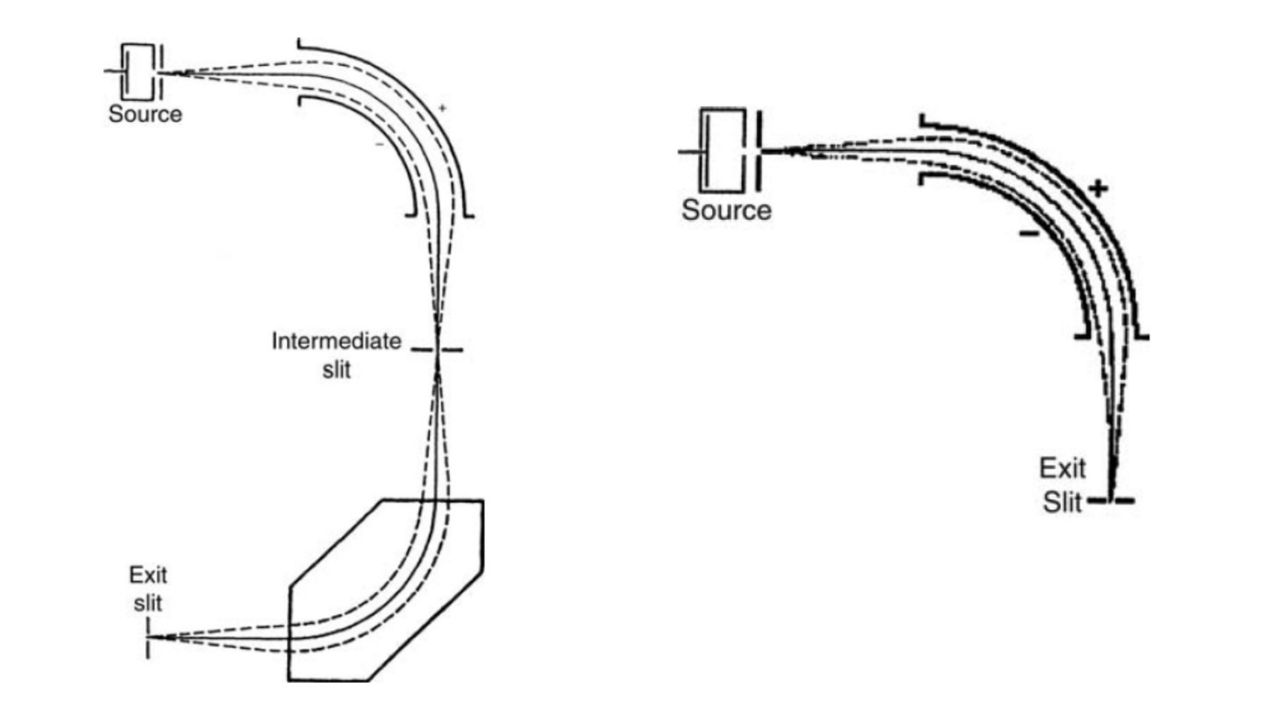
(d) Bainbridge, 1933

DISADVANTAGE

- The ions are formed from molecules that have a Boltzmann distribution of energies to begin with.
- This results in a distribution of velocities and adversely affects the resolution of the instrument by broadening the signal at the detector.
- The ion source has small variations in its electric field gradient, causing ions formed in different regions of the source to experience different acceleration.

ELECTRIC SECTOR

- The electrostatic field is an electric sector and separates ions by kinetic energy, not by mass.
- The ion beam from the source can be made much more homogeneous with respect to velocities of the ions if the beam is passed through an electric sector before being sent to the mass analyzer.
- The electric sector acts as an energy filter; only ions with a very narrow kinetic energy distribution will pass through.



ELECTRIC SECTOR

- The electrostatic field is an electric sector and separates ions by kinetic energy, not by mass.
- The ion beam from the source can be made much more homogeneous with respect to velocities of the ions if the beam is passed through an electric sector before being sent to the mass analyzer.
- The electric sector acts as an energy filter; only ions with a very narrow kinetic energy distribution will pass through.

ELECTRIC SECTOR

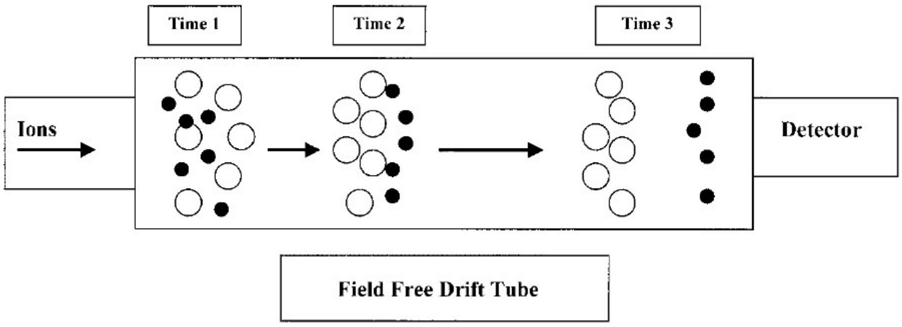
- The electrostatic field is an electric sector and separates ions by kinetic energy, not by mass.
- The ion beam from the source can be made much more homogeneous with respect to velocities of the ions if the beam is passed through an electric sector before being sent to the mass analyzer.
- The electric sector acts as an energy filter; only ions with a very narrow kinetic energy distribution will pass through.

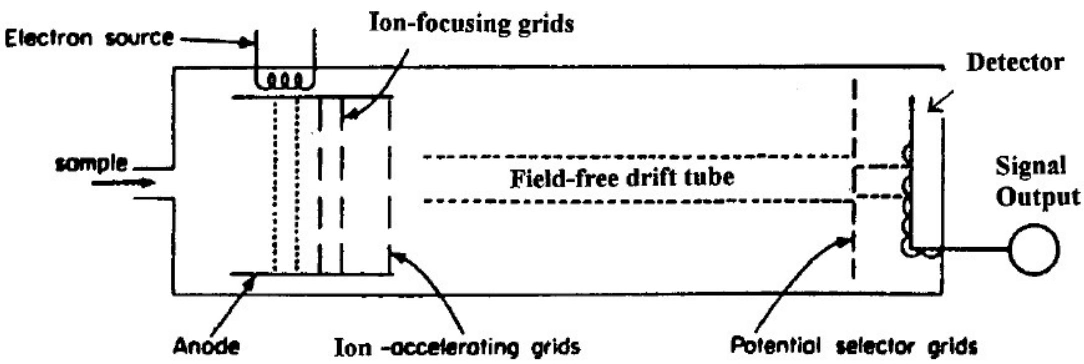
Time of Flight (TOF) Analyzer

- Pulses of ions are accelerated into an evacuated field free region called a drift tube.
- If all ions have the same kinetic energy, then the velocity of an ion depends on its mass-to-charge ratio or on its mass, if all ions have the same charge.
- Lighter ions will travel faster along the drift tube than heavier ions and are detected first.
- The drift tube in a TOF system is approximately 1–2 m in length.
- Pulses of ions are produced from the sample using pulses of electrons, secondary ions, or laser pulses.
- Ion pulses are produced with frequencies of 10–50 kHz.

Time of Flight (TOF) Analyzer

- The ions are accelerated into the drift tube by a pulsed electric field, called the ion-extraction field, because it extracts (or draws out) ions into the field-free region.
- Accelerating voltages up to 30 kV and extraction pulse frequencies of 5–20 kHz are used.





Time of Flight (TOF) Analyzer

Ions are separated in the drift tube according to their velocities. The velocity of an ion, v, can be expressed as:

$$v = \sqrt{\frac{2zV}{m}}$$

where V is the accelerating voltage. If L is the length of the field-free drift tube and t is the time from acceleration to detection of the ion (i.e., the flight time of the ion in the tube),

$$v = \frac{L}{t}$$

Time of Flight (TOF) Analyzer

and the equation that describes ion separation is:

$$\frac{m}{z} = \frac{2Vt^2}{L^2}$$

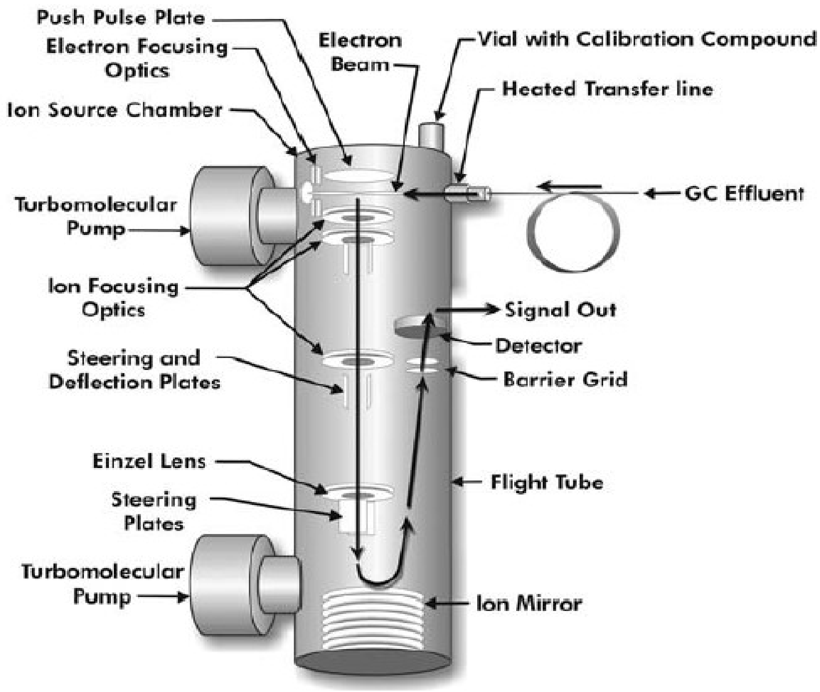
The flight time, t, of an ion is:

$$t = L\sqrt{\frac{m}{2zV}}$$

Time of Flight (TOF) Analyzer

- TOF instruments were first developed in the 1950s, but fell out of use because of the inherent low resolution of the straight drift tube design.
- The drift tube length and flight time are fixed, so resolution depends on the accelerating pulse.
- Ion pulses must be kept short to avoid overlap of one pulse with the next, which would cause mass overlap and decrease resolution.
- Interest in TOF instruments resurfaced in the 1990s with the introduction of MALDI and rapid data acquisition methods.
- The simultaneous transmission of all ions and the rapid flight time means that the detector can capture the entire mass spectral range almost instantaneously.

- The resolution of a TOF analyzer can be enhanced by the use of an ion mirror, called a reflectron.
- The reflectron is used to reverse the direction in which the ions are traveling and to energy-focus the ions to improve resolution.
- The reflectron's electrostatic field allows faster ions to penetrate more deeply than slower ions of the same m/z value. The faster ions follow a longer path before they are turned around, so that ions with the same m/z value but differing velocities end up traveling exactly the same distance and arrive at the detector together.
- In a reflectron TOF, the ion source and the detector are at the same end of the spectrometer; the reflectron is at the opposite end from the ion source.
- The ions traverse the drift tube twice, moving from the ion source to the reflectron and then back to the detector.

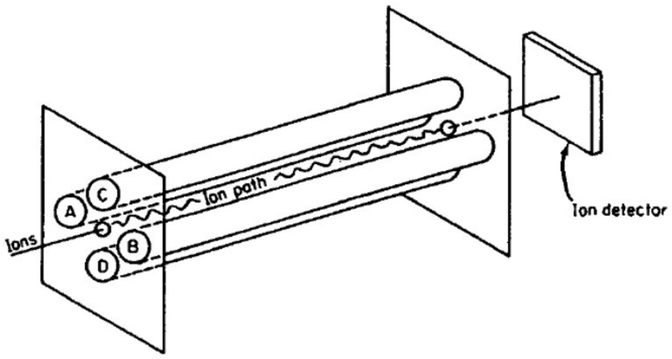


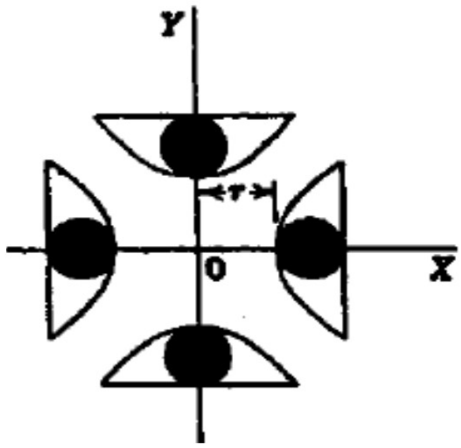
Quadrupole Mass Analyzer

- The quadrupole separates ions in an electric field (the quadrupole field) that is varied with time.
- This field is created using an oscillating radio frequency (RF) voltage and a constant direct current (DC) voltage applied to a set of four precisely machined parallel metal rods.
- This results in an AC potential superimposed on the DC potential.
- The ion beam is directed axially between the four rods.
- The opposite pairs of rods A and B, and C and D, are each connected to the opposite ends of a DC source, such that when C and D are positive, A and B are negative.
- The pairs of electrodes are then connected to an electrical source oscillating at RFs.

Quadrupole Mass Analyzer

- They are connected in such a way that the potentials of the pairs are continuously 180^o out of phase with each other.
- The magnitude of the oscillating voltage is greater than that of the DC source, resulting in a rapidly oscillating field.
- The RF voltage can be up to 1200 V while the DC voltage is up to 200 V.





Quadrupole Mass Analyzer

- An ion introduced into the space between the rods is subjected to a complicated lateral motion due to the DC and RF fields.
- Assume that the x direction is the line through the midpoint of the cross-sections of rods A and B; the y direction is the line through the midpoint of the cross-sections of rods C and D.
- The motion is complex because the velocity in the x direction is a function of the position along y and vice versa.

$$\frac{d^2x}{dt^2} + \frac{2}{r^2(m/z)}(V_{DC} + V_{RF}\cos 2\pi ft)y = 0$$

$$\frac{d^2y}{dt^2} + \frac{2}{r^2(m/z)}(V_{DC} + V_{RF}\cos 2\pi ft)x = 0$$

Quadrupole Mass Analyzer

- In order for an ion to pass through the space between the four rods, every time a positive ion is attracted to a negatively charged rod, the AC electric field must be present to push it away; otherwise, it will collide with the rod and be lost.
- The coordination between the oscillating (AC) field and the time of the ion's arrival at a rod surface over the fixed distance between the rods is critical to an ion's movement through the quadrupole.
- As a result of being alternately attracted and repelled by the rods, the ions follow an oscillating or "corkscrew" path through the quadrupole to the detector.
- <u>For a given amplitude of a fixed ratio of DC to RF at a fixed frequency, only ions of a given m/z value will pass through the quadrupole</u>

DETECTION OF ION

- ✓ DC voltage
- ✓ AC voltage
- ✓ Frequency of oscillation of AC
- ✓ m/z
- ❖ If the mass-to-charge ratio of the ion and the frequency of oscillation fit in above mentioned equations, the ion will oscillate toward the detector and eventually reach.
- ❖ Only a single m/z value can pass through the quadrupole at a given set of conditions. In this respect, the quadrupole acts like a filter, and is often called a mass filter.

METHODS FOR THE SEPARATION OF IONS

- ➢ The frequency of oscillation of the RF field can be held constant while varying the potentials of the DC and RF fields in such a manner that their ratio(0.168) is kept constant.
- ➢ varying the frequency of Ac field while kept the the DC and AC fields voltage constant.
- ➢ resolution of the system is dependent

1. number of oscillations an ion
2. Increasing the rod lengths
3. frequency of the RF field

- ❖ If the diameter is increased, the sensitivity is greatly increased, but then the mass range of the system is decreased.

MS-MS AND MS^N INSTRUMENTS(tandem in space)